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Course: CSA0421 - Operating System | **GitHub Repo:** krishshiva721/CSA04--Assignment1

TASK 9: MEMORY HIERARCHY & I/O ARCHITECTURE

Data Placement Strategy and Direct Memory Access (DMA) Transfer

1. MEMORY HIERARCHY & HOSPITAL DATA MAPPING

Hierarchy Level	Access Time	Typical Capacity	Assigned Hospital Data Component
CPU Registers	< 0.5 ns	< 2 KB	Active sensor readings, current alarm thresholds, loop counters.
L1 / L2 / L3 Cache	1 – 12 ns	4 MB – 128 MB	Hot patient lookup index structures, database transaction locks.
Main Memory (DRAM)	50 – 100 ns	16 GB – 512 GB	Active EHR transaction sessions, 3D MRI reconstruction buffer arrays.
NVMe SSD Array	20 – 100 µs	2 TB – 50 TB	Central active PostgreSQL database files, high-speed write ahead logs.
Network SAN / NAS	2 – 10 ms	Petabytes	Long-term historical DICOM MRI image archives, partner hospital shares.

2. OPTIMAL I/O METHOD SELECTION FOR DEVICES

MRI Imaging Server (Direct Memory Access - DMA): High-throughput volumetric MRI scans transfer hundreds of megabytes per second. Using DMA allows the optical image acquisition card to write data streams directly into main memory DRAM over PCI Express lanes, bypassing CPU registers entirely and freeing compute cores for GPU reconstruction calculations.

3. ALTERNATIVE CONCEPT & MATHEMATICAL OVERHEAD ANALYSIS

> **Alternative Scenario: Interrupt-Driven I/O vs. DMA for 500 MB MRI Image Processing**
>
> Consider transferring a single 500 MB 3D MRI image slice array (524,288,000 bytes) using 64-bit word words (8 bytes per word transfer):
>
> - **Interrupt-Driven I/O Overhead:** Total interrupts generated $N = 524,288,000 / 8 = 65,536,000$ interrupts. If each interrupt context switch consumes 1.5 microseconds, total CPU overhead consumed solely on interrupt handling equals $65.536 \times 10^6 \times 1.5 \times 10^{-6}$ s $= 98.3$ seconds of 100% CPU lockup!
> - **DMA Transfer Overhead:** Generates exactly **1 single interrupt** upon complete transfer of the 500 MB block. CPU intervention time is reduced to < 5 µs (a 19,000,000× overhead reduction).